

SISTEM INFORMASI MANAJEMEN EVENT ALPHA.CORP

Ringkasan Project

Disusun untuk kebutuhan presentasi dan dokumentasi

Kelompok 6 – PBL 2D

Politeknik Negeri Padang

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1. Gambaran Umum Project

Alpha.corp adalah aplikasi berbasis web yang dibangun menggunakan framework Laravel 12 untuk mengelola layanan event secara terintegrasi. Sistem ini menghubungkan tiga aktor utama — Admin Event Organizer, Klien/Customer, dan Vendor/Mitra Event — dalam satu platform digital yang terpusat.

Tujuan Aplikasi

- Mempermudah pengelolaan layanan event secara digital dan terstruktur
- Mempercepat komunikasi dan koordinasi antara klien, admin, dan vendor
- Meningkatkan efisiensi penyusunan proposal, RAB, dan dokumen legal
- Memudahkan monitoring pelaksanaan event secara real-time
- Menyediakan dokumentasi dan laporan akhir event yang terpusat

Masalah yang Diselesaikan

- Proses pengelolaan event yang masih manual dan tidak terintegrasi
- Sulitnya memantau progres event secara real-time
- Dokumentasi event yang tidak terpusat dan rawan hilang
- Komunikasi antara pihak terkait yang kurang efektif
- Pembuatan proposal, invoice, dan laporan yang masih manual

Target Pengguna

- Admin Event Organizer
- Klien / Customer
- Vendor / Mitra Event

2. Tech Stack

Lapisan	Teknologi	Keterangan
Backend	Laravel 12 + PHP 8.2+	Framework MVC modern
Frontend	Blade + Bootstrap 5	Template engine + CSS framework
Database	MySQL	Relational database
PDF	Barryvdh/DomPDF 3.0	Generate proposal, invoice, kontrak, kwitansi
Excel	Maatwebsite/Excel 3.1	Export analytics multi-sheet (6 sheet)
Auth	Laravel Breeze 2.4 + Socialite 5.28	Auth scaffolding + Google OAuth
Image	Intervention/Image 4.1	Manipulasi gambar upload
Build	Vite + Tailwind CSS	Asset bundling & utility CSS
Testing	Pest PHP 3.8	Testing framework modern

3. Arsitektur Aplikasi

Aplikasi ini mengadopsi arsitektur berlapis (Layered Architecture) dengan Repository Pattern:

Route (web.php)

Mendefinisikan endpoint URL dan middleware role-based access



Controller

Menerima request, validasi (FormRequest), memanggil Service, mengembalikan view



Service (Business Logic)

Seluruh logika: create event, generate PDF, kirim email, negosiasi



Repository (Data Access)

Abstraksi query database via interface contract



Model (Eloquent ORM)

Representasi tabel + relasi, accessor, computed attribute



Database (MySQL)

30+ tabel berelasi

Dependency Injection: Controller → Service → Repository via constructor.

Repository Pattern: 10 interface di-binding ke implementasi di AppServiceProvider.

4. Alur Bisnis Utama

1. Klien mendaftar/login → Mengajukan kebutuhan event (form detail event)
2. Admin membuat event → Menunjuk PIC admin → Menyusun proposal & RAB
3. Admin generate Surat Penawaran (PDF) → Dikirim ke klien (notifikasi & email)
4. Klien lihat penawaran → Pilih: Setuju langsung ATAU Ajukan Negosiasi
5. Jika negosiasi: Klien ajukan revisi → Admin setuju/tolak → Kirim revisi penawaran
6. Klien setuju → Proposal status 'diterima' → Kontrak aktif
7. Admin generate Invoice (DP + Pelunasan) → Klien upload bukti bayar
8. Admin verifikasi pembayaran → Assign vendor ke event
9. Admin buat Timeline kegiatan & Tugas untuk vendor
10. Vendor lihat tugas → Update progress → Upload dokumentasi
11. Admin monitoring progress → Complete timeline → Generate Laporan Akhir (PDF)
12. Klien lihat laporan & dokumentasi → Beri feedback

Status Workflow Event

Status	Keterangan
menunggu	Pengajuan baru, belum diproses
diproses	Penyusunan proposal/negosiasi
berjalan	Event sedang berlangsung
selesai	Event selesai dilaksanakan
dibatalkan	Event dibatalkan

5. Struktur Direktori & Penjelasan per File

5.1 app/Models/ — 30 Model Eloquent

Setiap model merepresentasikan tabel database dengan relasi, accessor, dan helper method.

Model	Fungsi Utama
User	Multi-role (admin/client/vendor), relasi events, vendor, notifikasi, avatar, last_active_at
Event	Event utama. Relasi: client, PIC, vendors, proposals, kontrak, RAB, invoice, payment, timeline, dokumentasi. Computed: progress, total_invoice, sisa_tagihan
Vendor	Profil vendor, many-to-many ke Event via event_vendor pivot (jadwal, status, harga)
Proposal	Proposal/penawaran dengan versioning (versi), status workflow, is_active flag
Rab	Item RAB (nama_item, volume, satuan, harga_satuan, subtotal_biaya, vendor_id)
RabAdditionalDetail	Detail biaya tambahan RAB
Invoice	Invoice per event (bisa >1: DP + Pelunasan), status workflow
Payment	Riwayat pembayaran + bukti upload + status verifikasi
Contract	Kontrak kerja per event
Negotiation	Riwayat negosiasi admin-klien
Timeline	Kegiatan event (nama, tanggal, status, PIC, hasil)
Task	Tugas spesifik untuk vendor
Document / DocumentSend	Dokumen upload & riwayat pengiriman ke klien
Documentation / DocumentationFile	Dokumentasi event + file pendukung
Feedback	Ulasan/feedback klien
Notification / AdminClientNotification	Notifikasi sistem & notifikasi admin ke klien
PaymentScheme	Skema pembayaran (DP, termin)
EventVendor	Pivot event-vendor (jadwal, status, harga, deskripsi)
Portfolio/Service/Team/Client	Data CMS landing page
LandingSection/AboutSection/AboutStatistic	Konten landing page
Report	Laporan akhir event

5.2 app/Http/Controllers/

Admin Controllers

DashboardController	Dashboard: total event, klien, vendor, revenue, pending tasks
EventController	CRUD event + filter/search/pagination
ProposalController	Upload dokumen, kirim/revisi penawaran (PDF), negosiasi
RabController	Manajemen RAB (item biaya per event)
VendorController	CRUD vendor + assign ke event
EventVendorController	Atur jadwal, status & harga vendor per event
TimelineController	Atur timeline kegiatan
TaskController	Atur tugas vendor
PaymentController	Verifikasi pembayaran, generate invoice & kwitansi PDF
KelolaKlienController	Kelola akun client (aktif/nonaktif, kirim notif)
ClientRequestController	Lihat & kelola pengajuan event klien
NotificationController	Kirim notifikasi massal ke klien
DocumentationController	Atur dokumentasi event
DocumentBuilderController	Builder dokumen PDF
AnalyticsController	Laporan analitik (chart, statistik, export Excel)
SettingsController	Pengaturan sistem
CmsController	CMS landing page (services, portfolio, team, clients)

Auth Controllers

- AuthenticatedSessionController
- RegisteredUserController
- GoogleController
- PasswordController
- NewPasswordController
- PasswordResetLinkController
- ConfirmablePasswordController
- EmailVerificationPromptController
- EmailVerificationNotificationController
- VerifyEmailController

Client Controller

ClientController — Dashboard, event, proposal (lihat/terima/negosiasi), invoice, payment upload, timeline, dokumentasi, feedback, notifikasi, settings.

Vendor Controllers

VendorController	Ringkasan, event saya, pengaturan
TugasController	Lihat & update status tugas
DokumentasiController	Upload dokumentasi event

VendorController	Ringkasan, event saya, pengaturan
NotifikasiController	Lihat & tandai baca notifikasi

Public Controllers

- LandingPageController
- CompanyProfileController
- FeedbackController
- ProfileController

5.3 app/Services/ — Business Logic Layer (25+ Service)

Memisahkan logika bisnis dari controller agar lebih testable dan mudah dipelihara.

Service	Fungsi
AdminDashboardService	Statistik dashboard: total event, klien, vendor, revenue, pending tasks
AdminEventService	CRUD event, filter/search, pagination
AdminProposalService	Logic proposal: generate PDF surat penawaran, kirim email, versioning, negosiasi, revisi
AdminVendorService	CRUD vendor, assign ke event, filter
AdminPaymentService	Generate invoice & kwitansi PDF, verifikasi pembayaran
AdminTaskService	CRUD tugas vendor
AdminDocumentationService	Kelola dokumentasi event
AdminNotificationService	Kirim notifikasi massal ke klien
AdminAnalyticsService	Data analitik: revenue per bulan/tahun, event stats, chart
AdminSettingsService	Pengaturan sistem
ClientService	Dashboard client, proposal, negosiasi, payment, timeline, notif
ClientRequestService	Pengajuan kebutuhan event oleh klien
RabService	Perhitungan & manajemen RAB
TimelineService / TimelineAutoFill	Manajemen timeline & auto-fill milestone
VendorDashboardService	Ringkasan vendor: event, tugas, event terdekat
CmsService / LandingPageService	CRUD CMS landing page & data publik
CompanyProfileService	Generate company profile PDF real-time
PaymentSchemeService	Manajemen skema pembayaran
DocumentBuilderService	Builder dokumen PDF kustom
EventVendorService	Logic assignment vendor ke event

5.4 app/Repositories/ & app/Interfaces/

10 interface (contract) di-binding ke repository implementasi di AppServiceProvider. Repository mengabstraksi query database.

Interface	Repository	Fungsi
DocumentRepositoryInterface	DocumentRepository	Query dokumen, pagination, filter tipe
ProposalRepositoryInterface	ProposalRepository	Versioning proposal, count harian, negosiasi
VendorRepositoryInterface	VendorRepository	Query vendor, filter event
EventVendorRepositoryInterface	EventVendorRepository	Query pivot event-vendor
TaskRepositoryInterface	TaskRepository	Query tugas vendor
RabRepositoryInterface	RabRepository	Query RAB & total biaya
RabAdditionalDetailRepositoryInterface	RabAdditionalDetailRepository	Query detail tambahan RAB
TimelineRepositoryInterface	TimelineRepository	Query timeline
NotificationRepositoryInterface	NotificationRepository	Query notifikasi
FeedbackRepositoryInterface	FeedbackRepository	Query feedback

Binding dilakukan di AppServiceProvider.register() dengan method ->app->bind().

5.5 app/Exports/ — Export Excel Analytics

Menggunakan Maatwebsite/Excel 3.1 dengan multi-sheet export (6 sheet):

File Export	Isi Sheet
AnalyticsExport (master)	Menggabungkan 6 sheet dalam satu file Excel
AnalyticsSummarySheet	Ringkasan statistik: total event, revenue, klien, vendor
AnalyticsEventsSheet	Detail events: nama, client, PIC, tanggal, status, progress
AnalyticsInvoicesSheet	Data invoice: nomor, event, nilai, status, tanggal
AnalyticsPaymentsSheet	Data pembayaran: invoice, nominal, status, metode, tanggal
AnalyticsClientsSheet	Data klien: nama, email, phone, total event, status
AnalyticsVendorsSheet	Data vendor: nama, jenis, alamat, total nilai, event aktif

5.6 app/Providers/ & app/Http/Middleware/

Providers

- AppServiceProvider — Binding Interface → Repository + share unread notif count ke semua view

Middleware

Middleware	Fungsi
AdminMiddleware	Cek user login & role='admin', abort 403 jika bukan
vendor.role (inline)	Cek role vendor di route group vendor

5.7 database/migrations/ — 39 File Migration

Urutan pembuatan tabel:

- Batch 1: Auth & Cache — users, cache, jobs
- Batch 2: Master Data — vendors, events, event_vendor, proposals, contracts
- Batch 3: Keuangan — rabs, invoices, payments
- Batch 4: Manajemen — timelines, documentations, reports, documentation_files
- Batch 5: Landing Page — landing_page_tables, cms_content_tables
- Batch 6: Notifikasi & Task — notifications, tasks, jadwal, feedback, documents, document_sends
- Batch 7: Negosiasi — negotiations
- Batch 8: Alter & Penambahan Kolom — 12 migration untuk relasi, status_pembayaran, RAB additional, payment schemes, dll

5.8 resources/views/ — Blade Templates

6 layout utama + puluhan halaman view per role akses:

Folder	Fungsi
layouts/	6 layout: app, admin, client, vendor, guest, landing
landing/	Hero, about, visi-misi, why-us, services, team, portfolio, clients
auth/	Login, register, forgot/reset password, verifikasi email
profile/	Edit profil, update password, delete akun
admin/dashboard/	Dashboard admin dengan statistik
admin/events/	CRUD event (index, form, show)
admin/proposals/	Dokumen, invoice PDF, proposal PDF
admin/rab/	Manajemen RAB
admin/timeline/	Timeline kegiatan
admin/payments/	Verifikasi pembayaran
admin/requests/	Surat penawaran, negosiasi
admin/kelola_klien/	Manajemen akun client
admin/cms/	CMS: clients, portfolio, services, team
admin/analytics/	Analitik & laporan PDF
admin/pdf_templates/	Template PDF: invoice, kwitansi, proposal, RAB, kontrak
client/	Dashboard, events, proposals, invoices, feedback, timeline, notif, settings
vendor/	Ringkasan, event-saya, daftar-tugas, notifikasi, pengaturan
pdf/ / emails/	Company profile PDF & template email notifikasi

5.9 routes/web.php

File routing utama (+- 400 baris) yang mendefinisikan seluruh endpoint:

Group	Prefix	Middleware	Endpoint
Public	/	—	Landing page, company profile PDF
Auth		guest	Login, register, Google OAuth, reset password
Admin	admin	auth+admin	Dashboard, events, proposals, RAB, vendor, payment, CMS, analytics
Client		auth+verified	Event, proposals, negosiasi, invoices, feedback, timeline, notif
Vendor	vendor	auth+vendor	Ringkasan, event-saya, daftar-tugas, dokumentasi, notif

Setiap route group dilindungi middleware spesifik sesuai role akses masing-masing.

5.10 tests/ — Testing (Pest PHP)

40+ file test mencakup Unit, Feature, dan Service test:

Folder	Isi
Unit/Models	EventTest, ProposalTest, TimelineTest, UserTest, VendorTest
Unit/Services	AdminVendorServiceTest, ClientServiceTest, EventVendorServiceTest, RabServiceTest, TimelineAutoFillTest
Unit/Repositories	EventVendorRepositoryTest, VendorRepositoryTest
Unit/Auth	AuthControllersTest, GoogleControllerTest, LoginRequestTest
Feature/Admin	RAB, Task, Vendor management
Feature/Auth	Auth, registration, password, email verification
Feature/Client	Access control, dashboard, invoice, proposal, settings
Feature/Vendor	Vendor feature tests

Jalankan: `composer test` atau spesifik: `composer test:admin`, `composer test:services`, dll.

5.11 File Konfigurasi & Pendukung

File	Fungsi
composer.json	Dependency: Laravel 12, DomPDF, Socialite, Intervention, Maatwebsite Excel
package.json	Dependency: Vite, Tailwind CSS, Alpine.js, Axios
.env / .env.example	Konfigurasi environment (DB, mail, Google OAuth)
config/dompdf.php	Konfigurasi DomPDF (paper size, orientation, font)
config/excel.php	Konfigurasi Maatwebsite Excel
config/app.php / auth.php / database.php	Konfigurasi aplikasi, auth, & database
config/mail.php / session.php	Konfigurasi mail server & session
config/filesystems.php / services.php	Filesystem disk & third-party services
vite.config.js / tailwind.config.js	Vite bundler & Tailwind CSS config
phpunit.xml	Konfigurasi PHPUnit/Pest
docs/	Panduan instalasi, testing, dependencies
README.md	Dokumentasi project lengkap
ActivityDiagram.xml	Diagram aktivitas (draw.io)

6. Fitur per Role

Fitur Admin

- Dashboard real-time dengan statistik
- Manajemen Event (CRUD, filter, search)
- Manajemen Proposal & Surat Penawaran (PDF, email)
- Manajemen RAB (item biaya, subtotal)
- Manajemen Vendor & Assignment ke Event
- Manajemen Timeline & Tugas Vendor
- Verifikasi Pembayaran & Generate Invoice PDF
- Negosiasi dengan Klien (setujui/tolak/revisi)
- Kelola Akun Klien (aktif/nonaktif, kirim notifikasi)
- CMS Landing Page (services, portfolio, team, clients)
- Dokumentasi Event & Laporan Akhir
- Export Analytics Excel (6 sheet)
- Generate Company Profile PDF
- Pengaturan Sistem

Fitur Client

- Dashboard pribadi daftar event
- Pengajuan kebutuhan event baru
- Lihat proposal & surat penawaran (PDF preview & download)
- Setujui penawaran langsung
- Ajukan negosiasi (dengan catatan & file)
- Lihat invoice & upload bukti pembayaran
- Lihat timeline event
- Lihat dokumentasi event
- Lihat laporan akhir event
- Beri feedback & rating
- Notifikasi real-time
- Pengaturan akun (profil, password)

Fitur Vendor

- Ringkasan (event, tugas aktif/selesai, event terdekat)
- Event Saya (daftar event di-assign)
- Daftar Tugas & Update Status
- Upload Dokumentasi Event
- Notifikasi Sistem

- Pengaturan Akun

7. Diagram & Alur Sistem

File ActivityDiagram.xml (draw.io) dapat dibuka untuk diagram aktivitas sistem secara visual.

Relasi Database

Users (1) → (N) Events (client_id / pic_admin_id)
Events (N) ↔ (N) Vendors (via event_vendor pivot)
Events (1) → (N) Proposals (versioning)
Events (1) → (N) RABs (item biaya)
Events (1) → (N) Invoices → (N) Payments
Events (1) → (N) Timelines (kegiatan)
Events (1) → (N) Tasks (tugas vendor)
Events (1) → (N) Documentations → (N) DocumentationFiles
Events (1) → (N) Negotiations
Events (1) → (1) Contracts
Events (1) → (N) Feedbacks
Users (1) → (N) Notifications
Vendors (1) → (N) RABs (vendor_id)

Alur Request-Response

Request → Route → Middleware → Controller (FormRequest) → Service (logic) → Repository (data) → Model → Database → Response

Dokumen ini digenerate secara otomatis dari kode sumber project.
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